

Selecting a Water-Wave Theory

How to read the Le Méhauté (1976) region-of-validity diagram and choose between linear, Stokes, cnoidal and solitary wave theories. Companion note to the online Wave Theory Selection calculator.

1. Why theory selection matters

No single closed-form wave theory is accurate over the whole range of water depths and wave heights. Linear (Airy) theory assumes an infinitesimally small wave height; real waves have finite height, so their crests are sharper and troughs flatter than a sine curve, and they transport mass and momentum that linear theory omits. Higher-order Stokes theories correct for finite height in deep and transitional water, while cnoidal and solitary theories describe the long, peaked waves found in shallow water. Choosing the wrong theory can badly misestimate wave kinematics, forces and run-up. The Le Méhauté diagram is the standard chart engineers use to make this choice.

2. Governing dimensionless numbers

Three dimensionless quantities control which theory applies. They are all obtained once the wavelength L is found from the linear dispersion relation (solved iteratively):

$$\omega^2 = g \cdot k \cdot \tanh(kd), \quad \omega = 2\pi/T, \quad k = 2\pi/L, \quad L = L_0 \cdot \tanh(kd), \quad L_0 = gT^2/2\pi.$$

Relative depth d/L measures how much the bed ‘feels’ the wave and defines the depth regime. **Wave steepness** H/L measures nonlinearity in deep water; it cannot exceed about 0.142 before the wave breaks. The **Ursell number** combines the two and is the key discriminator between Stokes-type and cnoidal waves:

$$U_R = H \cdot L^2 / d^3. \quad U_R < 26 \rightarrow \text{Stokes / Airy family}; \quad U_R > 26 \rightarrow \text{cnoidal theory}.$$

3. Water-depth regimes

Regime	Relative depth d/L	Behaviour
Deep water	$d/L \geq 0.5$	Waves do not feel the bed; $L \rightarrow L_0$; orbits are closed circles.
Transitional	$0.05 < d/L < 0.5$	Both depth and period matter; the working range of most coastal sites.
Shallow water	$d/L \leq 0.05$	Strongly depth-controlled; $c \rightarrow \sqrt{gd}$; orbits flatten to ellipses.

4. The wave theories in brief

Linear (Airy): first-order, sinusoidal; valid for small steepness in any depth — the everyday engineering default. **Stokes 2nd–5th order:** perturbation series in wave steepness for deep and transitional water; successive orders sharpen the crest and are needed as steepness rises. **Cnoidal:** based on Jacobian elliptic (cn) functions for long, finite-height waves in shallow water ($U_R > 26$); reduces to linear for low waves and to the solitary wave in the long-wave limit. **Solitary:** a single translatory crest of positive elevation, the limiting shallow-water wave approached near breaking.

5. The Le Méhauté (1976) diagram

The chart plots the wave-height parameter H/gT^2 against the relative-depth parameter d/gT^2 on logarithmic axes. Because d/gT^2 fixes d/L through the dispersion relation, lines of constant d/L are vertical: the deep-water ($d/L=0.5$) and shallow-water ($d/L=0.05$) limits appear as the two dashed vertical lines. The upper envelope is the breaking limit; the dashed curve is the Ursell $U_R=26$ boundary separating the Stokes/Airy family (lower-right) from cnoidal waves (upper-left). Plotting a wave's (d/gT^2 , H/gT^2) point on the chart immediately shows which region — and therefore which theory — applies.

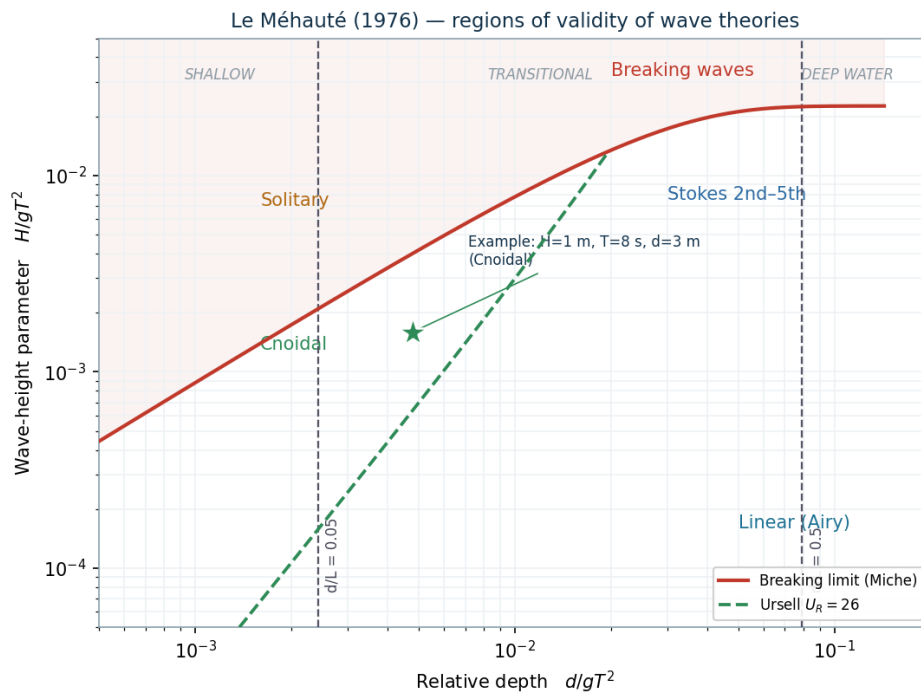


Figure 1. Regions of validity of wave theories after Le Méhauté (1976). The star is the worked example of Section 7 ($H=1$ m, $T=8$ s, $d=3$ m), which falls in the cnoidal region.

6. Breaking limits

The tool checks the Miche (1944) steepness limit, which spans all depths and reduces to the familiar depth-limited criterion in shallow water:

$$H_b/L = 0.142 \cdot \tanh(kd) \quad (\text{Miche}); \quad \text{deep water: } H_b/L_0 \approx 0.142; \quad \text{shallow water: } H_b/d \approx 0.78\text{--}0.83.$$

If the input height reaches H_b , the wave is breaking and no periodic theory is valid; the calculator reports the ratio H/H_b as a breaking-proximity check.

7. Selection procedure and worked example

The calculator applies the following logic: if $H \geq H_b$ the wave is breaking; else if $U_R \geq 26$ and $d/L < 0.5$ it is cnoidal (solitary when H/d is large); otherwise it is a Stokes wave whose order rises with steepness, reducing to linear (Airy) theory at low steepness. This reproduces the region boundaries of Figure 1.

Quantity	Value
Inputs	$H = 1.0$ m, $T = 8.0$ s, $d = 3.0$ m
Deep-water length, $L_0 = gT^2/2\pi$	99.93 m
Wavelength, L (dispersion)	42.03 m
Relative depth, d/L	0.0714 (transitional)
Wave steepness, H/L	0.0238
Ursell number, $U_R = HL^2/d^3$	65.4
Chart point (d/gT^2 , H/gT^2)	(0.00478, 0.00159)
Miche breaking height, H_b	2.51 m (wave at 40% of H_b)
Recommended theory	Cnoidal ($U_R > 26$, not breaking)

Notation

H wave height; T period; d water depth; L wavelength; L_0 deep-water wavelength; $k = 2\pi/L$ wavenumber; $\omega = 2\pi/T$ angular frequency; g gravitational acceleration; U_R Ursell number; H_b breaking wave height.

References

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